

Key points

- A buffer of wealth protects households against bad income draws.
- The endogenous grid method solves the model without root finding.
- Patience, returns and risk set the size of the buffer.

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Correspondence

Jane Economist
jane@example.org

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Buffer Stock Saving with Heterogeneous Agents

An example of the Econ-ARK MyST template

Jane Economist^{*1}  , and John Modeler^{1,2†}

¹Department of Economics, Johns Hopkins University , ²Econ-ARK; [†]Contributed Equally

Abstract We solve a buffer stock saving model with the endogenous grid method and show how the Econ-ARK template typesets a working paper: title block, margin notes, equations, tables, code and author-year citations.

Summary Households save more when their income is uncertain. This paper shows how much more, and how quickly the endogenous grid method computes it.

Keywords consumption, precautionary saving, heterogeneous agents

JEL codes C63, D14, E21

Materials

Run online
Launch
Starts in a few minutes

Code
[econ-ark/econ-ark-myst](#)
MIT license

REMARK
[BufferStockTheory](#)

Source
[github.com/econ-ark/econ-ark-myst](#)

Also as
Latest version

For everyone who waited for the solver to converge.

All models are wrong, but some are useful. George Box

1 Introduction

Precautionary saving arises when income risk interacts with a convex marginal utility (Carroll 1997). The endogenous grid method (Carroll 2006) makes such models cheap to solve, because it replaces root finding with an inversion of the Euler equation.¹ Households hold a buffer of wealth against bad income draws (Carroll 1997; 2006). The size of that buffer depends on patience, the return on saving and the degree of risk.

2 Model

The consumer maximizes expected discounted utility subject to the budget constraint

^{*}Also a visiting scholar at Econ-ARK.

¹The method inverts the first-order condition on a grid of end-of-period assets rather than solving it on a grid of market resources. `IndShockConsumerType` applies it to Section 2's model.

$$v(m_t) = \max_{c_t} u(c_t) + \beta \mathbb{E}_t[v(m_{t+1})], \quad m_{t+1} = R(m_t - c_t) + y_{t+1}. \quad (1)$$

(1) implies the Euler equation $u'(c_t) = \beta R \mathbb{E}_t[u'(c_{t+1})]$. Under constant relative risk aversion, $u(c) = c^{1-\rho}/(1-\rho)$.

2.1 Calibration

Table 1 Baseline calibration.

Parameter	Description	Value
β	Discount factor	0.96
R	Gross interest factor	1.03
ρ	Relative risk aversion	2.00

2.1.1 Sources of the parameters

Both the discount factor and the interest factor follow [Carroll \(1997\)](#), while we set risk aversion to the value the literature uses most often.

3 Results

Table 1 lists the parameters used in the solution, which come from the 1990's literature. Proposition 1 describes the shape of the solution, and Figure 1 shows it at that calibration.

Proposition 1 (Concavity). *If income risk is present, the consumption function is strictly concave. Its slope falls toward the perfect foresight marginal propensity to consume as wealth grows.*

Proof. See [Carroll \(1997\)](#) for the argument under constant relative risk aversion. $\square$

Note

Admonitions take a rule in the palette rather than the filled box MyST draws by default. Econ-ARK blue carries this kind and important.

Tip

Green carries tip, hint and seealso.

Warning

Orange carries attention, caution and this kind.

Danger

Pink carries this kind and error.



Figure 1 Solution at the calibration of Table 1, over the same range of market resources.

Buffer stock Wealth an impatient consumer facing income risk holds against a bad draw, toward which wealth returns from either side.

Perfect foresight Section 2's problem with income risk removed, which Carroll (1997) treats as the limiting case.

Prudence and impatience together pin down a target level of wealth, which the consumer saves toward from below and spends down toward from above.

```
from HARK.ConsumptionSaving.ConsIndShockModel import
IndShockConsumerType

agent = IndShockConsumerType() # the 1990's calibration of Table 1
agent.solve()
```

Appendix A derives the Euler equation.

Declarations

The authors report no competing interests. The authors used a large language model to draft the example text of this template and reviewed every sentence before release.

Acknowledgments

We thank the Econ-ARK community for comments.

Data availability

All code needed to reproduce the results is in the repository linked in the margin.

References

- Carroll, Christopher D. 1997. "Buffer-Stock Saving and the Life Cycle/Permanent Income Hypothesis." *The Quarterly Journal of Economics* 112 (1): 1–56. <https://doi.org/10.1162/003355397555109>.
- Carroll, Christopher D. 2006. "The Method of Endogenous Gridpoints for Solving Dynamic Stochastic Optimization Problems." *Economics Letters* 91 (3): 312–20. <https://doi.org/10.1016/j.econlet.2005.09.013>.

Appendix A Derivation of the Euler equation

The first-order condition for c_t in (1) sets $u'(c_t)$ equal to $\beta R \mathbb{E}_t[v'(m_{t+1})]$. The envelope condition gives $v'(m_t) = u'(c_t)$, and substituting it yields the Euler equation.